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Division of Electronics and Communication Engineering
2025-2026 (EVEN SEM)

III IA EVALUATION REPORT

for

24EC2006-DIGITAL SIGNAL PROCESSINGPROJECT BASED COURSE

Title of the project: Digital Audio Noise Reduction using FIR and IIR Filtering Techniques

A report submitted by

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Subject Code	24EC2006
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Project Rubrics for Evaluation

First Review: Project title selection - PPT should have four slides (Title page, Introduction, Circuit/Block Diagram, and Description of Project).

Second Review: PPT should have three slides (Description of Concept, implementation, outputs, results and discussion)

Rubrics for project (III IA - 40 Marks):
Content - 4 marks (based on Project)
Clarity - 3 marks (based on viva during presentation)
Feasibility - 3 marks (based on project)
Presentation - 10 marks
Project Report - 10 marks
On-time submission - 5 marks (before the due date)
Online submission-GCR - 5 marksTotal marks: _____/ 40 Marks

Signature of Faculty with date:

TABLE OF CONTENTS

CHAPTER	TITLE	PAGE NO.
1.	INTRODUCTION	3
2.	DESCRIPTION OF THE PROJECT	4
3.	CONCEPT INVOLVED	5-6
4.	TOOLS	7
5.	IMPLEMENTATION	8
6.	RESULTS WITH GRAPH/SIMULATION	9-11
7.	INFERENCES	12
8.	CONCLUSION	13

CHAPTER 1

INTRODUCTION

In modern digital signal processing, audio signals are often affected by unwanted noise during recording, transmission, or storage. This noise degrades the quality and intelligibility of the original signal, making noise reduction an essential requirement in various applications such as communication systems, music processing, and speech analysis.

Digital Signal Processing (DSP) techniques are widely used to enhance and restore audio signals by removing unwanted disturbances. Among these techniques, digital filtering plays a crucial role in isolating useful signal components from noise based on their frequency characteristics.

Key Concepts

- **Audio Noise**
  Unwanted disturbances that affect the quality of the original signal
- **Digital Filtering**
  Technique used to remove noise and improve signal clarity
- **FIR (Finite Impulse Response) Filter**
  A stable filter with linear phase response, suitable for precise signal processing
- **IIR (Infinite Impulse Response) Filter**
  An efficient filter that requires less computation and provides faster processing

Digital filters such as FIR and IIR are widely used due to their effectiveness in noise reduction and signal enhancement. Each filter has its own advantages and limitations in terms of stability, complexity, and performance.

This project focuses on the use of FIR and IIR filters for improving audio signal quality using MATLAB, highlighting their importance in modern signal processing applications.

CHAPTER 2

DESCRIPTION OF THE PROJECT

This project deals with the practical implementation of digital signal processing techniques for reducing noise in audio signals using MATLAB. In many real-time applications, audio signals get distorted due to background noise, interference, and limitations of recording systems. Such distortions reduce the usability and clarity of the signal, making noise reduction an important task.

The main objective of this project is to simulate a noisy environment and apply suitable digital filtering techniques to recover the original signal. A clean audio signal is first considered and noise is artificially introduced to replicate real-world conditions. This allows better analysis of how effectively different filters can remove unwanted components.

Two important types of digital filters, namely FIR (Finite Impulse Response) and IIR (Infinite Impulse Response), are designed and implemented. These filters are applied to the noisy signal, and their outputs are studied to understand their behavior, efficiency, and performance in noise reduction.

- **Signal Preparation:** The input audio signal is loaded into MATLAB and converted into a suitable format for processing.

- **Noise Simulation:** Random noise is added to the signal to create a noisy environment for testing.

- **FIR Filter Application:** The FIR filter is implemented to reduce noise while maintaining stability and linear phase response.

- **IIR Filter Application:** The IIR filter is applied to achieve efficient noise reduction with lower computational complexity.

- **Output Analysis:** The filtered signals are observed and compared to evaluate improvement in quality.

- **Performance Comparison:** The efficiency, stability, and output characteristics of both filters are compared.

CHAPTER 3

CONCEPT INVOLVED

This project is based on the fundamental concepts of Digital Signal Processing (DSP) applied to audio signals and noise reduction using digital filtering techniques. The important concepts involved are explained below:

1. Digital Audio Signal

An audio signal represents sound in digital form, obtained by sampling a continuous-time signal at a specific sampling frequency (fs). In this project, the signal is treated as a discrete-time signal for processing in MATLAB.

2. Noise in Signals

Noise is an unwanted disturbance that affects the original signal during recording or transmission. In this project, Gaussian random noise is artificially added to simulate real-world noisy conditions and test filter performance.

3. FIR (Finite Impulse Response) Filter

An FIR filter produces an output that depends only on present and past input values.
• Always stable in nature
• Provides linear phase response
• Designed using window method (fir1 function)
FIR filters are suitable for applications where signal stability and waveform preservation are important.

4. IIR (Infinite Impulse Response) Filter

An IIR filter generates output based on both input and previous output values.
• Requires less computation compared to FIR
• Achieves sharper filtering with lower order
In this project, a Butterworth filter is used for efficient noise removal within a specific frequency range.

5. Filtering Process

Filtering is used to remove unwanted frequency components from a signal.
• Low-pass filtering removes high-frequency noise
• Bandpass filtering retains only required frequency components

6. Time Domain Analysis

In time domain analysis, signals are observed with respect to time (amplitude vs time).
This helps in understanding the effect of noise addition and the improvement after filtering.

7. Frequency Domain Analysis (FFT)

FFT (Fast Fourier Transform) converts signals from time domain to frequency domain.
It helps to:
• Identify noise frequency components
• Compare signal spectrum before and after filtering

8. Signal Playback

Audio playback is used to hear the difference between original, noisy, and filtered signals.
This provides a practical understanding of how filtering improves audio quality.

CHAPTER 4

TOOLS

The following tools and software are used in this project for the design, implementation, and analysis of audio noise reduction using digital filters:

1. MATLAB Software

MATLAB is the main tool used for implementing the project. It provides a powerful environment for numerical computation, signal processing, and visualization. All filtering operations, signal analysis, and plotting are performed using MATLAB.

2. Signal Processing Toolbox

This toolbox provides built-in functions required for designing and analyzing digital filters. Functions such as fir1, butter, filter, freqz, and fft are used for implementing FIR and IIR filters and analyzing their performance.

3. Audio Processing Functions

MATLAB provides functions like audioread, sound, and audiowrite for handling audio signals. These functions are used to load, play, and save audio signals during the project.

4. FFT (Fast Fourier Transform)

FFT is used to convert signals from time domain to frequency domain. It helps in analyzing frequency components of the signal and identifying noise before and after filtering.

5. Graphical Visualization Tools

Plotting functions such as plot, subplot, and title are used to visualize signals in time domain and frequency domain. These graphs help in comparing original, noisy, and filtered signals.

6. Computer System

A computer or laptop is used to run MATLAB and execute the project. It provides the necessary platform for simulation and analysis of digital signals.

CHAPTER 5

IMPLEMENTATION

The project is implemented using MATLAB by processing an audio signal, adding noise, and applying FIR and IIR filters to observe the results through graphs and audio playback. The entire process is executed step-by-step using MATLAB commands, and the outputs are visualized for analysis.

Initially, a sample audio signal is loaded using MATLAB's built-in dataset. The signal is then converted into a time-domain representation for plotting. Random noise is added to the signal to simulate real-world noisy conditions.

- **Loading the Signal:**
  The command load handel is used to load a sample audio signal. The signal and sampling frequency are stored and prepared for processing.
- **Adding Noise:**
  Random noise is generated using randn() and added to the original signal to create a noisy signal.
- **FIR Filtering:**
  The FIR filter is designed using fir1() and applied using the filter() function to reduce noise.
- **IIR Filtering:**
  The IIR filter is designed using the butter() function and applied using filter() for efficient noise removal.
- **Graphical Output:**
  The signals are plotted using plot() and subplot() functions. Four graphs are displayed:
  → Original signal
  → Noisy signal
  → FIR filtered signal
  → IIR filtered signal
- **Frequency Response:**
  The freqz() command is used to display the frequency response of both FIR and IIR filters.
- **Command Window Output:**
  Messages like "Playing Original…", "Playing FIR Output…" are displayed using disp() to indicate processing stages.
- **Audio Playback:**
  The sound() function is used to play original, noisy, and filtered signals, allowing practical comparison.

CHAPTER 6

RESULTS WITH GRAPH/SIMULATION

The results of the project are obtained by simulating the audio signal processing in MATLAB. The performance of FIR and IIR filters is analyzed using graphical representation and audio playback. The results clearly show the improvement in signal quality after filtering.

Time Domain Results
The signals are plotted in the time domain using MATLAB plotting functions.

- **Original Signal:**
  Shows the clean audio waveform without any noise
- **Noisy Signal:**
  Displays the signal after adding random noise, showing distortions
- **FIR Filter Output:**
  The waveform becomes smoother with reduced noise and preserved shape
- **IIR Filter Output:**
  Noise is reduced effectively with faster response, but slight variations may be observed

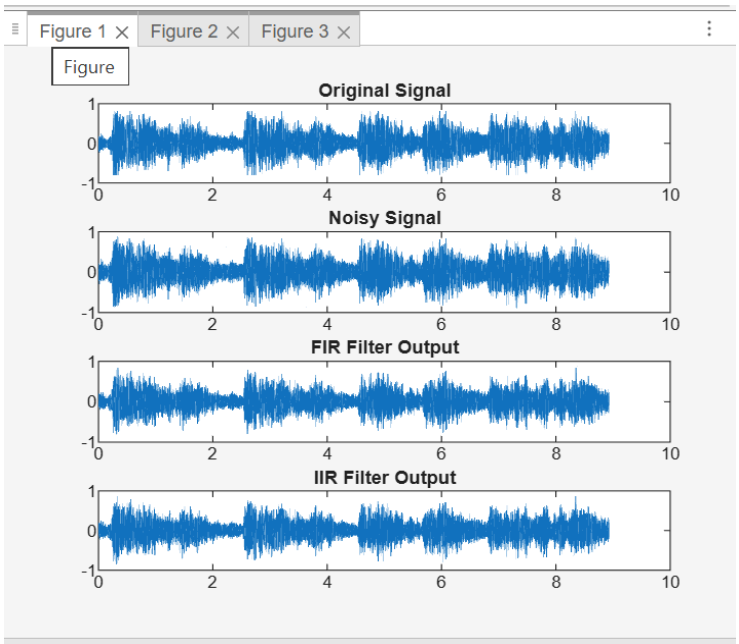
Frequency Domain Results
The frequency characteristics of the filters are analyzed using:

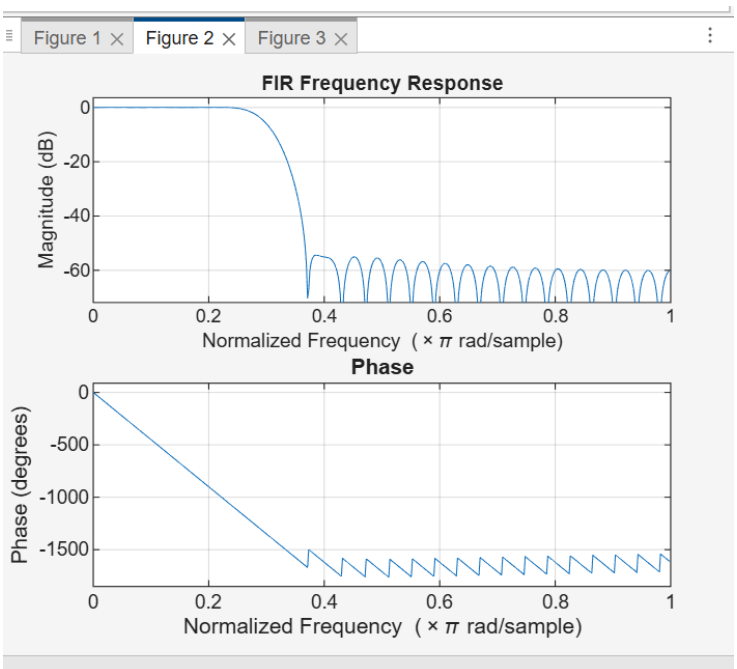
- **FIR Frequency Response (freqz):**
  Shows smooth attenuation of unwanted frequencies
- **IIR Frequency Response (freqz):**
  Displays sharper cutoff and efficient filtering
- **FFT Analysis:**
  Used to compare frequency components before and after filtering
  → Noise components are reduced after filtering

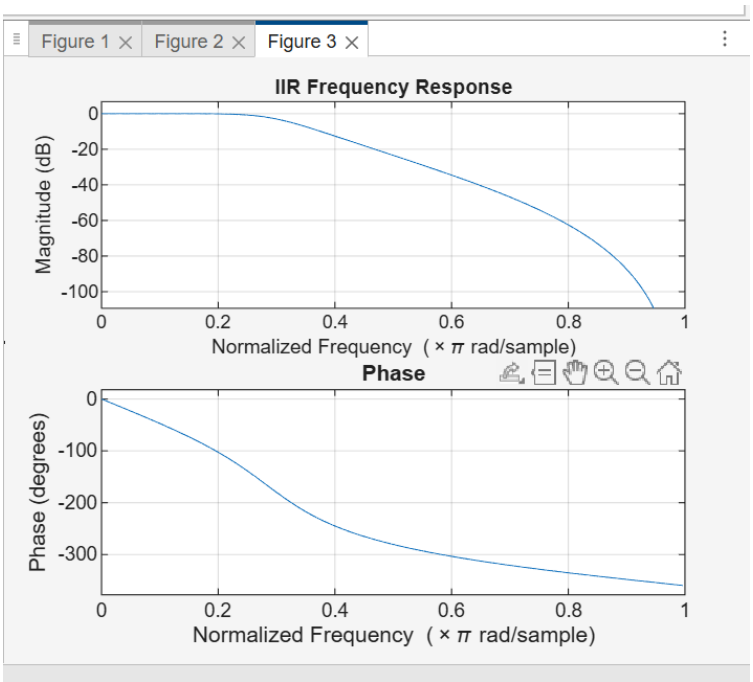
Simulation Output
- MATLAB successfully generates multiple plots using subplot()
- Clear comparison between original, noisy, and filtered signals
- Frequency response graphs validate filter design
- Output audio signals are saved using audiowrite()

Observations
- Noise is clearly visible in the noisy signal and reduced after filtering
- FIR filter provides smoother and more stable output
- IIR filter provides faster and efficient noise removal
- Both filters significantly improve audio signal quality

The simulation results confirm that digital filters are effective in reducing noise from audio signals, and the graphical outputs help in clearly visualizing the improvements achieved.







Command Window

New to MATLAB? See resources for Getting Started.

```
Playing Original...
Playing Noisy...
Playing FIR Output...
Playing IIR Output...
Program Executed Successfully!
>> Press Ctrl + Shift + P to generate code with Copilot
```

CHAPTER 7

INFERENCES

- The presence of noise significantly degrades the quality and clarity of audio signals
- Digital filtering techniques are effective in removing unwanted noise components
- FIR filters provide stable and smooth output with linear phase characteristics
- IIR filters achieve faster noise reduction with lower computational complexity
- FIR filters require higher order compared to IIR filters for similar performance
- IIR filters may introduce slight distortion due to non-linear phase response
- Time-domain and frequency-domain analysis help in clearly understanding filter performance
- MATLAB provides an efficient platform for implementing and analyzing DSP techniques

CHAPTER 8

CONCLUSION

This project successfully demonstrates the application of digital signal processing techniques for reducing noise in audio signals using MATLAB. By implementing both FIR and IIR filters, the project clearly shows how unwanted noise can be effectively removed to improve the clarity and quality of the original signal.

The analysis of results indicates that FIR filters provide excellent stability and linear phase response, which helps in preserving the original waveform without distortion. On the other hand, IIR filters offer faster processing and require less computational effort, making them suitable for real-time applications.

The graphical analysis in both time domain and frequency domain, along with audio playback, confirms the effectiveness of the filtering techniques. It also highlights the trade-offs between FIR and IIR filters in terms of performance, complexity, and efficiency.

Overall, this project provides a strong understanding of digital filtering concepts and their practical implementation, proving that FIR and IIR filters are essential tools for audio noise reduction in modern signal processing systems.